

the Fast Fourier Transform

- 1 Time and Frequency Domains
 - fitting periodic data
- 2 the Fast Fourier Transform
 - via the butterfly matrix
 - the recursion
- 3 Filter Design
 - convolutions with the FFT

MCS 472 Lecture 10
Industrial Math & Computation
Jan Verschelde, 4 February 2026

the Fast Fourier Transform

1 Time and Frequency Domains

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fitting periodic data

Data is often periodic, e.g.: seasonal temperature fluctuations.

To fit periodic data, a basis of cosines and sines works well.

Given (t_i, f_i) , $i = 0, 1, \dots, n$, compute the coefficients of

$$y(t) = c_0 + c_1 \cos(2\pi t) + c_2 \sin(2\pi t),$$

so $y(t)$ fits the data (t_i, f_i) best.

A Fourier series approximation is written as

$$y(t) = a_0 + \sum_{k=0}^n (a_k \cos(2\pi kt) + b_k \sin(2\pi kt)).$$

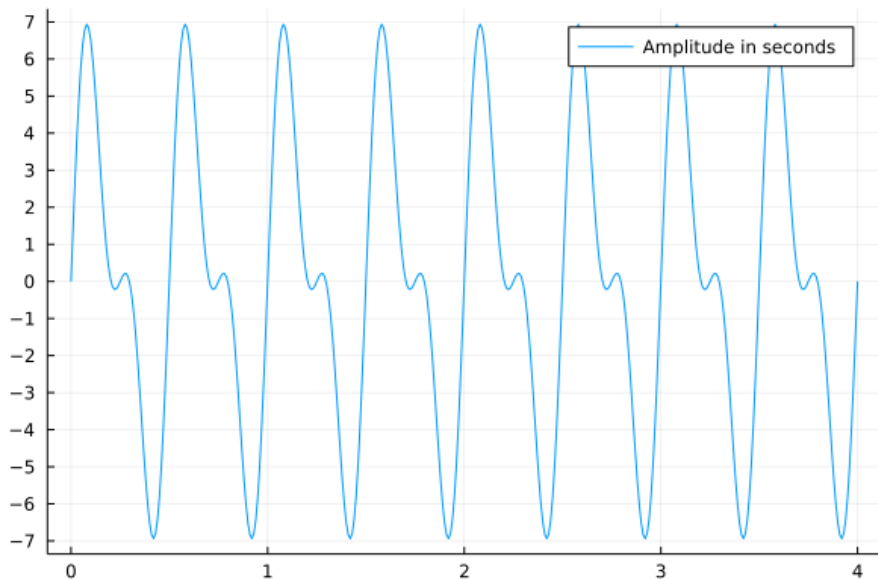
The Fast Fourier Transform solves the interpolation problem in time $O(n \log(n))$ for n data points.

an experiment in Julia

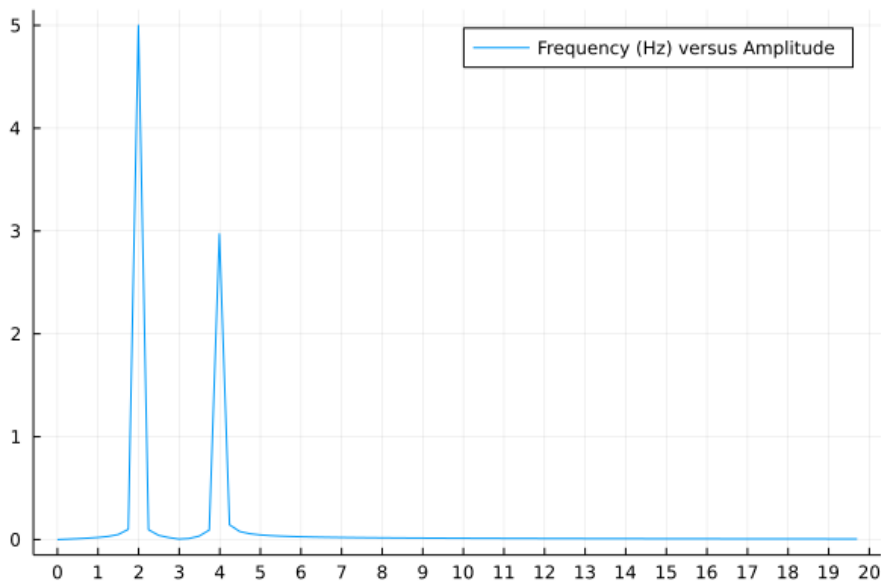
```
using Plots
using FFTW

dt = 0.01
t = 0:dt:4
# amplitudes 3 and 5 at frequencies 4 and 2
y = 3*sin.(4*2*pi*t) + 5*sin.(2*2*pi*t)
plot(t, y, yticks=-7:1:7,
      label="Amplitude in seconds")
F = fft(y)
n = length(y)/2
amps = abs.(F)/n
freq = [0:79]/(2*n*dt)
plot(freq, amps[1:80], xticks=0:1:20,
      label="Frequency (Hz) versus Amplitude")
```

amplitude versus time



frequency versus amplitude



verify the output of `FFTW.fft`

Exercise 1:

Use your `FourierMatrix` of Exercise 2 of Lecture 9 to compare the outcome of $F_n \mathbf{x}$ computed with the F_n computed with your `FourierMatrix` and the output of `FFTW.fft`.

Do the comparison on a random vector $\mathbf{x}$ of size $n = 8$.

Does the `FFTW.fft` use the normalization factor $1/\sqrt{n}$?

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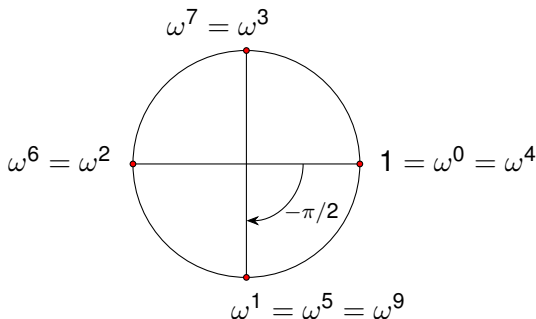
3 Filter Design

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the Fourier matrix

For $n = 4$, consider $\hat{\mathbf{x}} = \frac{1}{\sqrt{4}} M_4 \mathbf{x}$, with $\omega = e^{-i2\pi/4}$ and

$$M_4 = \begin{bmatrix} \omega^0 & \omega^0 & \omega^0 & \omega^0 \\ \omega^0 & \omega^1 & \omega^2 & \omega^3 \\ \omega^0 & \omega^2 & \omega^4 & \omega^6 \\ \omega^0 & \omega^3 & \omega^6 & \omega^9 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & \omega & \omega^2 & \omega^3 \\ 1 & \omega^2 & 1 & \omega^2 \\ 1 & \omega^3 & \omega^2 & \omega \end{bmatrix}$$



permute and partition

$$M_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & \omega & \omega^2 & \omega^3 \\ 1 & \omega^2 & 1 & \omega^2 \\ 1 & \omega^3 & \omega^2 & \omega \end{bmatrix} \quad P = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Multiplying M_4 with P swaps columns 2 and 3.

$$M_4 P = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & \omega^2 & \omega & \omega^3 \\ 1 & 1 & \omega^2 & \omega^2 \\ 1 & \omega^2 & \omega^3 & \omega \end{bmatrix} = \left[\begin{array}{cc|cc} 1 & 1 & 1 & 1 \\ 1 & \omega^2 & \omega & \omega^3 \\ 1 & 1 & \omega^2 & \omega^2 \\ 1 & \omega^2 & \omega^3 & \omega \end{array} \right]$$

Observe the repeated occurrence of M_2 :

$$M_2 = \begin{bmatrix} 1 & 1 \\ 1 & \omega^2 \end{bmatrix}, \quad F_2 = \frac{1}{\sqrt{2}} M_2.$$

the matrix D_2 to multiply with ω

$$M_4 P = \left[\begin{array}{cc|cc} 1 & 1 & 1 & 1 \\ 1 & \omega^2 & \omega & \omega^3 \\ \hline 1 & 1 & \omega^2 & \omega^2 \\ 1 & \omega^2 & \omega^3 & \omega \end{array} \right], \quad M_2 = \begin{bmatrix} 1 & 1 \\ 1 & \omega^2 \end{bmatrix}, \quad F_2 = \frac{1}{\sqrt{2}} M_2.$$

Multiplying the second row by ω of M_2 , via a matrix multiplication:

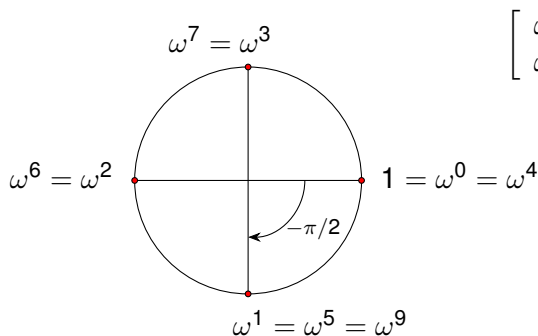
$$\begin{bmatrix} 1 & 0 \\ 0 & \omega \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & \omega^2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ \omega & \omega^3 \end{bmatrix} = D_2 M_2,$$

where D_2 is the multiplier matrix:

$$D_2 = \begin{bmatrix} 1 & 0 \\ 0 & \omega \end{bmatrix}.$$

flipping signs

$$M_4 P = \left[\begin{array}{cc|cc} 1 & 1 & 1 & 1 \\ 1 & \omega^2 & \omega & \omega^3 \\ \hline 1 & 1 & \omega^2 & \omega^2 \\ 1 & \omega^2 & \omega^3 & \omega \end{array} \right], \quad M_2 = \begin{bmatrix} 1 & 1 \\ 1 & \omega^2 \end{bmatrix}, \quad D_2 = \begin{bmatrix} 1 & 0 \\ 0 & \omega \end{bmatrix}.$$



$$\begin{bmatrix} \omega^2 & \omega^2 \\ \omega^3 & \omega \end{bmatrix} = \begin{bmatrix} -1 & -1 \\ -\omega & -\omega^3 \end{bmatrix} \\ = -D_2 M_2$$

the butterfly matrix

With

$$M_2 = \begin{bmatrix} 1 & 1 \\ 1 & \omega^2 \end{bmatrix} \quad \text{and} \quad D_2 = \begin{bmatrix} 1 & 0 \\ 0 & \omega \end{bmatrix}$$

we rewrite $M_4 P$ as

$$\left[\begin{array}{cc|cc} 1 & 1 & 1 & 1 \\ 1 & \omega^2 & \omega & \omega^3 \\ \hline 1 & 1 & \omega^2 & \omega^2 \\ 1 & \omega^2 & \omega^3 & \omega \end{array} \right] = \begin{bmatrix} M_2 & D_2 M_2 \\ M_2 & -D_2 M_2 \end{bmatrix} = \begin{bmatrix} I_2 & D_2 \\ I_2 & -D_2 \end{bmatrix} \begin{bmatrix} M_2 & 0 \\ 0 & M_2 \end{bmatrix}.$$

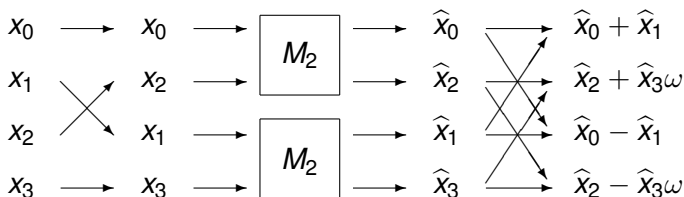
The matrix

$$B_4 = \begin{bmatrix} I_2 & D_2 \\ I_2 & -D_2 \end{bmatrix}, \quad I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

is called *the butterfly matrix*.

the butterfly picture

$$B_4 \begin{bmatrix} \hat{x}_0 \\ \hat{x}_2 \\ \hat{x}_1 \\ \hat{x}_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & \omega \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -\omega \end{bmatrix} \begin{bmatrix} \hat{x}_0 \\ \hat{x}_2 \\ \hat{x}_1 \\ \hat{x}_3 \end{bmatrix} = \begin{bmatrix} \hat{x}_0 + \hat{x}_1 \\ \hat{x}_2 + \hat{x}_3\omega \\ \hat{x}_0 - \hat{x}_1 \\ \hat{x}_2 - \hat{x}_3\omega \end{bmatrix}$$



verify for $n = 8$

Exercise 2:

Write a Julia function `ButterflyMatrix` which takes on input the dimension n and which returns the butterfly matrix of dimension n .

To verify the decomposition, define a function `mFourierMatrix` that takes on input the dimension n and which returns the Fourier matrix times $\sqrt{n}$.

For $n = 8$, verify the relationship between `mFourierMatrix( $n$ )` and `ButterflyMatrix( $n$ )`.

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the recursion

For any n , we apply the recursive factorization

$$M_n = B_n \begin{bmatrix} M_{n/2} & 0 \\ 0 & M_{n/2} \end{bmatrix} P_n$$

where the partition P_n reorders the even and odd indexed numbers, with the butterfly matrix

$$B_n = \begin{bmatrix} I_{n/2} & D_{n/2} \\ I_{n/2} & -D_{n/2} \end{bmatrix},$$

where $I_{n/2}$ is the identity matrix of dimension $n/2$, and $D_{n/2}$ is the diagonal matrix with on its diagonal

$$1, \omega, \omega^2, \dots, \omega^{n/2-1}.$$

$O(n \log(n))$ time

To the recurrence $T(n) \leq 2T(n/2) + O(n)$,
we apply the following theorem:

Theorem ($O(n \log(n))$ recurrence)

*If $T(n) \leq 2T(n/2) + c_1 n$ and $T(2) \leq c_2$,
for positive constants c_1 and c_2 , then $T(n)$ is $O(n \log(n))$.*

Exercise 3:

Prove the above theorem.

You may apply `rsolve` of SymPy.

the FFT runs in quasi linear time

Because $\log(n)$ grows so slowly, $O(n \log(n))$ is called *quasi linear time*.

Theorem (cost of FFT)

On vectors of size n , the FFT runs in $O(n \log(n))$ time.

For the inverse FFT, $F_n^{-1} = \overline{F}_n$, so

$$F_n^{-1} \mathbf{y} = \overline{F}_n \mathbf{y} = \overline{F_n \overline{\mathbf{y}}},$$

and no extra work is needed.

Exercise 4: Prove $\overline{F}_n \mathbf{y} = \overline{F_n \overline{\mathbf{y}}}$.

Does $\overline{F}_n \mathbf{y} = \overline{F_n \overline{\mathbf{y}}}$ hold for any complex matrix A ? Is $\overline{A} \mathbf{y} = \overline{A \overline{\mathbf{y}}}$?

verify the $O(n \log(n))$ time

Exercise 5:

Generate random vectors for size 2^k , for $k = 10, 11, \dots, 20$.

For each random vector, record the time for `FFTW.fftw`.

Do you observe the $O(n \log(n))$ time?

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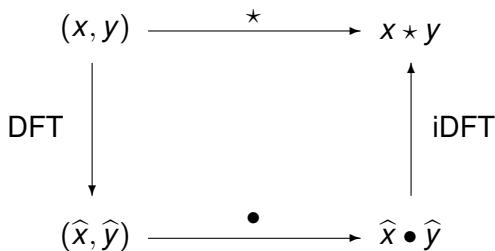
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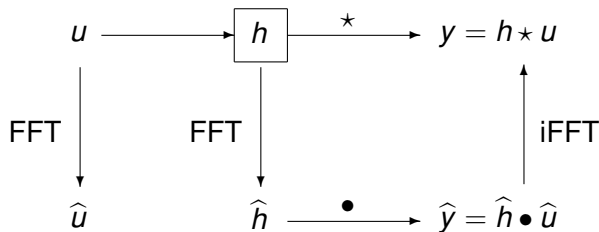
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the DFT convolution theorem



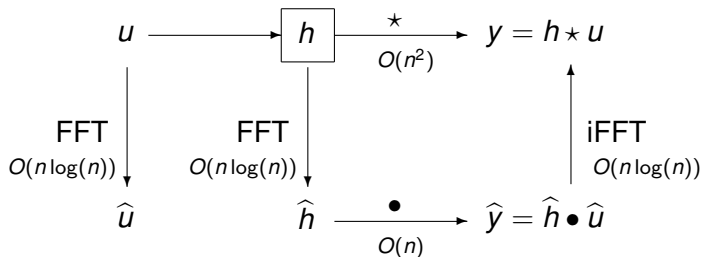
- The inverse discrete Fourier transform (iDFT)
- applied to the componentwise product $\hat{x} \bullet \hat{y}$
- of the discrete Fourier transforms (DFTs) $\hat{x}$ and $\hat{y}$,
- respectively of x and y , equals the convolution $x \star y$.

the convolution theorem applied to a filter with the FFT



With the FFT, the convolution of two n -vectors is $O(n \log(n))$.

the convolution theorem applied to a filter with the FFT



With the FFT, the convolution of two n -vectors is $O(n \log(n))$.

summary and bibliography

We defined the Fast Fourier Transform (FFT).

The main references for this lecture:

- Charles R. MacCluer:
Industrial Mathematics. Modeling in Industry, Science, and Government. Prentice Hall, 2000.

We are in the middle of Chapter 4.

- Timothy Sauer: *Numerical Analysis*,
second edition, Pearson, 2012.

Chapter 10 deals with the discrete Fourier transform.

The FFT is covered in most Algorithms courses, using as textbook *Algorithm Design*, by Jon Kleinberg and Éva Tardos, Pearson, 2006.